Setting to 5 frames

```
C:\Users\jamer\Desktop\412Proj4>java DemandPagingSimulator 5
Number of page frames set to 5.

Please choose from the following options:
0 - Exit
1 - Read reference string
2 - Generate reference string
3 - Display current reference string
4 - Simulate FIFO Algorithm
5 - Simulate OPT Algorithm
6 - Simulate LRU Algorithm
7 - Simulate LFU Algorithm
```

Option 1

```
1
Enter a series of numbers:
2 4 5 6 1 5 3 4 5 2 3 6 5 3 4 7 3 5 6
Valid ref.string: 2, 4, 5, 6, 1, 5, 3, 4, 5, 2, 3, 6, 5, 3, 4, 7, 3, 5, 6.
Please choose from the following options:
0 - Exit
1 - Read reference string
2 - Generate reference string
3 - Display current reference string
4 - Simulate FIFO Algorithm
5 - Simulate OPT Algorithm
6 - Simulate LRU Algorithm
7 - Simulate LFU Algorithm
```

Option 4

```
4
Memory Information:
Algorithm Type: FIFO
Length of Refernce String: 19
Number of Virtual Pages: 10
Number of Physical Pages: 5
---
Press enter to step through snapshots of physical memory
Or, enter "q" at any time to return to main menu.
Snapshot at step 1:
Program called virtual frame # 2
Physical frame 0:[2]
Physical frame 1:x
Physical frame 2:x
Physical frame 3:x
Physical frame 4:x
Page faults: Yes.
Victim frames: None.

Snapshot at step 2:
Program called virtual frame # 4
Physical frame 0: 2
Physical frame 1:[4]
Physical frame 2:x
Physical frame 3:x
Physical frame 4:x
Page faults: Yes.
Victim frames: None.

Snapshot at step 3:
Program called virtual frame # 5
Physical frame 0: 2
Physical frame 1: 4
Physical frame 2:[5]
Physical frame 3:x
Physical frame 4:x
Page faults: Yes.
Victim frames: None.

Snapshot at step 4:
Program called virtual frame # 6
Physical frame 0: 2
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3:[6]
Physical frame 4:x
Page faults: Yes.
Victim frames: None.

Snapshot at step 5:
Program called virtual frame # 1
Physical frame 0: 2
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3: 6
Physical frame 4:[1]
Page faults: Yes.
Victim frames: None.

Snapshot at step 6:
Program called virtual frame # 5
Physical frame 0: 2
Physical frame 1: 4
Physical frame 2:[5]
Physical frame 3: 6
Physical frame 4: 1
Page faults: No.
Victim frames: None.

Snapshot at step 7:
Program called virtual frame # 3
Physical frame 0:[3]
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3: 6
Physical frame 4: 1
Page faults: Yes.
Victim frames: 2

Snapshot at step 8:
Program called virtual frame # 4
Physical frame 0: 3
Physical frame 1:[4]
Physical frame 2: 5
Physical frame 3: 6
Physical frame 4: 1
Page faults: No.
Victim frames: None.

Snapshot at step 9:
Program called virtual frame # 5
Physical frame 0: 3
Physical frame 1: 4
Physical frame 2:[5]
Physical frame 3: 6
Physical frame 4: 1
Page faults: No.
Victim frames: None.

Snapshot at step 10:
Program called virtual frame # 2
Physical frame 0: 3
Physical frame 1:[2]
Physical frame 2: 5
Physical frame 3: 6
Physical frame 4: 1
Page faults: Yes.
Victim frames: 4

Snapshot at step 11:
Program called virtual frame # 3
Physical frame 0:[3]
Physical frame 1: 2
Physical frame 2: 5
Physical frame 3: 6
Physical frame 4: 1
Page faults: No.
Victim frames: None.

Snapshot at step 12:
Program called virtual frame # 6
Physical frame 0: 3
Physical frame 1: 2
Physical frame 2: 5
Physical frame 3:[6]
Physical frame 4: 1
Page faults: No.
Victim frames: None.

Snapshot at step 13:
Program called virtual frame # 5
Physical frame 0: 3
Physical frame 1: 2
Physical frame 2:[5]
Physical frame 3: 6
Physical frame 4: 1
Page faults: No.
Victim frames: None.

Snapshot at step 14:
Program called virtual frame # 3
Physical frame 0:[3]
Physical frame 1: 2
Physical frame 2: 5
Physical frame 3: 6
Physical frame 4: 1
Page faults: No.
Victim frames: None.

Snapshot at step 15:
Program called virtual frame # 4
Physical frame 0: 3
Physical frame 1: 2
Physical frame 2:[4]
Physical frame 3: 6
Physical frame 4: 1
Page faults: Yes.
Victim frames: 5

Snapshot at step 16:
Program called virtual frame # 7
Physical frame 0: 3
Physical frame 1: 2
Physical frame 2: 4
Physical frame 3:[7]
Physical frame 4: 1
Page faults: Yes.
Victim frames: 6

Snapshot at step 17:
Program called virtual frame # 3
Physical frame 0:[3]
Physical frame 1: 2
Physical frame 2: 4
Physical frame 3: 7
Physical frame 4: 1
Page faults: No.
Victim frames: None.

Snapshot at step 18:
Program called virtual frame # 5
Physical frame 0: 3
Physical frame 1: 2
Physical frame 2: 4
Physical frame 3: 7
Physical frame 4:[5]
Page faults: Yes.
Victim frames: 1

Snapshot at step 19:
Program called virtual frame # 6
Physical frame 0:[6]
Physical frame 1: 2
Physical frame 2: 4
Physical frame 3: 7
Physical frame 4: 5
Page faults: Yes.
Victim frames: 3
Simluation finished. Press enter to continue.
```

Option 5

```
5
Memory Information:
Algorithm Type: OPT
Length of Refernce String: 19
Number of Virtual Pages: 10
Number of Physical Pages: 5
---
Press enter to step through snapshots of physical memory
Or, enter "q" at any time to return to main menu.
```

```
Snapshot at step 1:
Program called virtual frame # 2
Physical frame 0:[2]
Physical frame 1:x
Physical frame 2:x
Physical frame 3:x
Physical frame 4:x
Page faults: Yes.
Victim frames: None.

Snapshot at step 2:
Program called virtual frame # 4
Physical frame 0: 2
Physical frame 1:[4]
Physical frame 2:x
Physical frame 3:x
Physical frame 4:x
Page faults: Yes.
Victim frames: None.

Snapshot at step 3:
Program called virtual frame # 5
Physical frame 0: 2
Physical frame 1: 4
Physical frame 2:[5]
Physical frame 3:x
Physical frame 4:x
Page faults: Yes.
Victim frames: None.

Snapshot at step 4:
Program called virtual frame # 6
Physical frame 0: 2
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3:[6]
Physical frame 4:x
Page faults: Yes.
Victim frames: None.

Snapshot at step 5:
Program called virtual frame # 1
Physical frame 0: 2
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3: 6
Physical frame 4:[1]
Page faults: Yes.
Victim frames: None.

Snapshot at step 6:
Program called virtual frame # 5
Physical frame 0: 2
Physical frame 1: 4
Physical frame 2:[5]
Physical frame 3: 6
Physical frame 4: 1
Page faults: No.
Victim frames: None.

Snapshot at step 7:
Program called virtual frame # 3
Physical frame 0: 2
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3: 6
Physical frame 4:[3]
Page faults: Yes.
Victim frames: 1
```

```
Snapshot at step 8:
Program called virtual frame # 4
Physical frame 0: 2
Physical frame 1:[4]
Physical frame 2: 5
Physical frame 3: 6
Physical frame 4: 3
Page faults: No.
Victim frames: None.

Snapshot at step 9:
Program called virtual frame # 5
Physical frame 0: 2
Physical frame 1: 4
Physical frame 2:[5]
Physical frame 3: 6
Physical frame 4: 3
Page faults: No.
Victim frames: None.

Snapshot at step 10:
Program called virtual frame # 2
Physical frame 0:[2]
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3: 6
Physical frame 4: 3
Page faults: No.
Victim frames: None.

Snapshot at step 11:
Program called virtual frame # 3
Physical frame 0: 2
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3: 6
Physical frame 4:[3]
Page faults: No.
Victim frames: None.

Snapshot at step 12:
Program called virtual frame # 6
Physical frame 0: 2
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3:[6]
Physical frame 4: 3
Page faults: No.
Victim frames: None.

Snapshot at step 13:
Program called virtual frame # 5
Physical frame 0: 2
Physical frame 1: 4
Physical frame 2:[5]
Physical frame 3: 6
Physical frame 4: 3
Page faults: No.
Victim frames: None.

Snapshot at step 14:
Program called virtual frame # 3
Physical frame 0: 2
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3: 6
Physical frame 4:[3]
Page faults: No.
Victim frames: None.
```

```
Snapshot at step 15:
Program called virtual frame # 4
Physical frame 0: 2
Physical frame 1:[4]
Physical frame 2: 5
Physical frame 3: 6
Physical frame 4: 3
Page faults: No.
Victim frames: None.

Snapshot at step 16:
Program called virtual frame # 7
Physical frame 0:[7]
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3: 6
Physical frame 4: 3
Page faults: Yes.
Victim frames: 2

Snapshot at step 17:
Program called virtual frame # 3
Physical frame 0: 7
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3: 6
Physical frame 4:[3]
Page faults: No.
Victim frames: None.

Snapshot at step 18:
Program called virtual frame # 5
Physical frame 0: 7
Physical frame 1: 4
Physical frame 2:[5]
Physical frame 3: 6
Physical frame 4: 3
Page faults: No.
Victim frames: None.

Snapshot at step 19:
Program called virtual frame # 6
Physical frame 0: 7
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3:[6]
Physical frame 4: 3
Page faults: No.
Victim frames: None.
Simluation finished. Press enter to continue.
```

Option 6

```
6
Memory Information:
Algorithm Type: LRU
Length of Refernce String: 19
Number of Virtual Pages: 10
Number of Physical Pages: 5
---
Press enter to step through snapshots of physical memory
Or, enter "q" at any time to return to main menu.
```

```
Snapshot at step 1:
Program called virtual frame # 2
Physical frame 0:[2]
Physical frame 1:x
Physical frame 2:x
Physical frame 3:x
Physical frame 4:x
Page faults: Yes.
Victim frames: None.

Snapshot at step 2:
Program called virtual frame # 4
Physical frame 0: 2
Physical frame 1:[4]
Physical frame 2:x
Physical frame 3:x
Physical frame 4:x
Page faults: Yes.
Victim frames: None.

Snapshot at step 3:
Program called virtual frame # 5
Physical frame 0: 2
Physical frame 1: 4
Physical frame 2:[5]
Physical frame 3:x
Physical frame 4:x
Page faults: Yes.
Victim frames: None.

Snapshot at step 4:
Program called virtual frame # 6
Physical frame 0: 2
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3:[6]
Physical frame 4:x
Page faults: Yes.
Victim frames: None.

Snapshot at step 5:
Program called virtual frame # 1
Physical frame 0: 2
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3: 6
Physical frame 4:[1]
Page faults: Yes.
Victim frames: None.

Snapshot at step 6:
Program called virtual frame # 5
Physical frame 0: 2
Physical frame 1: 4
Physical frame 2:[5]
Physical frame 3: 6
Physical frame 4: 1
Page faults: No.
Victim frames: None.

Snapshot at step 7:
Program called virtual frame # 3
Physical frame 0:[3]
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3: 6
Physical frame 4: 1
Page faults: Yes.
Victim frames: 2
```

```
Snapshot at step 8:
Program called virtual frame # 4
Physical frame 0: 3
Physical frame 1:[4]
Physical frame 2: 5
Physical frame 3: 6
Physical frame 4: 1
Page faults: No.
Victim frames: None.

Snapshot at step 9:
Program called virtual frame # 5
Physical frame 0: 3
Physical frame 1: 4
Physical frame 2:[5]
Physical frame 3: 6
Physical frame 4: 1
Page faults: No.
Victim frames: None.

Snapshot at step 10:
Program called virtual frame # 2
Physical frame 0: 3
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3:[2]
Physical frame 4: 1
Page faults: Yes.
Victim frames: 6

Snapshot at step 11:
Program called virtual frame # 3
Physical frame 0:[3]
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3: 2
Physical frame 4: 1
Page faults: No.
Victim frames: None.

Snapshot at step 12:
Program called virtual frame # 6
Physical frame 0: 3
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3: 2
Physical frame 4:[6]
Page faults: Yes.
Victim frames: 1

Snapshot at step 13:
Program called virtual frame # 5
Physical frame 0: 3
Physical frame 1: 4
Physical frame 2:[5]
Physical frame 3: 2
Physical frame 4: 6
Page faults: No.
Victim frames: None.

Snapshot at step 14:
Program called virtual frame # 3
Physical frame 0:[3]
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3: 2
Physical frame 4: 6
Page faults: No.
Victim frames: None.
```

```
Snapshot at step 15:
Program called virtual frame # 4
Physical frame 0: 3
Physical frame 1:[4]
Physical frame 2: 5
Physical frame 3: 2
Physical frame 4: 6
Page faults: No.
Victim frames: None.

Snapshot at step 16:
Program called virtual frame # 7
Physical frame 0: 3
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3:[7]
Physical frame 4: 6
Page faults: Yes.
Victim frames: 2

Snapshot at step 17:
Program called virtual frame # 3
Physical frame 0:[3]
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3: 7
Physical frame 4: 6
Page faults: No.
Victim frames: None.

Snapshot at step 18:
Program called virtual frame # 5
Physical frame 0: 3
Physical frame 1: 4
Physical frame 2:[5]
Physical frame 3: 7
Physical frame 4: 6
Page faults: No.
Victim frames: None.

Snapshot at step 19:
Program called virtual frame # 6
Physical frame 0: 3
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3: 7
Physical frame 4:[6]
Page faults: No.
Victim frames: None.
Simluation finished. Press enter to continue.
```

Option 7

```
7
Memory Information:
Algorithm Type: LFU
Length of Refernce String: 19
Number of Virtual Pages: 10
Number of Physical Pages: 5
---
Press enter to step through snapshots of physical memory
Or, enter "q" at any time to return to main menu.
```

```
Snapshot at step 1:
Program called virtual frame # 2
Physical frame 0:[2]
Physical frame 1:x
Physical frame 2:x
Physical frame 3:x
Physical frame 4:x
Page faults: Yes.
Victim frames: None.

Snapshot at step 2:
Program called virtual frame # 4
Physical frame 0: 2
Physical frame 1:[4]
Physical frame 2:x
Physical frame 3:x
Physical frame 4:x
Page faults: Yes.
Victim frames: None.

Snapshot at step 3:
Program called virtual frame # 5
Physical frame 0: 2
Physical frame 1: 4
Physical frame 2:[5]
Physical frame 3:x
Physical frame 4:x
Page faults: Yes.
Victim frames: None.

Snapshot at step 4:
Program called virtual frame # 6
Physical frame 0: 2
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3:[6]
Physical frame 4:x
Page faults: Yes.
Victim frames: None.

Snapshot at step 5:
Program called virtual frame # 1
Physical frame 0: 2
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3: 6
Physical frame 4:[1]
Page faults: Yes.
Victim frames: None.

Snapshot at step 6:
Program called virtual frame # 5
Physical frame 0: 2
Physical frame 1: 4
Physical frame 2:[5]
Physical frame 3: 6
Physical frame 4: 1
Page faults: No.
Victim frames: None.

Snapshot at step 7:
Program called virtual frame # 3
Physical frame 0:[3]
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3: 6
Physical frame 4: 1
Page faults: Yes.
Victim frames: 2
```

```
Snapshot at step 8:
Program called virtual frame # 4
Physical frame 0: 3
Physical frame 1:[4]
Physical frame 2: 5
Physical frame 3: 6
Physical frame 4: 1
Page faults: No.
Victim frames: None.

Snapshot at step 9:
Program called virtual frame # 5
Physical frame 0: 3
Physical frame 1: 4
Physical frame 2:[5]
Physical frame 3: 6
Physical frame 4: 1
Page faults: No.
Victim frames: None.

Snapshot at step 10:
Program called virtual frame # 2
Physical frame 0:[2]
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3: 6
Physical frame 4: 1
Page faults: Yes.
Victim frames: 3

Snapshot at step 11:
Program called virtual frame # 3
Physical frame 0: 2
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3:[3]
Physical frame 4: 1
Page faults: Yes.
Victim frames: 6

Snapshot at step 12:
Program called virtual frame # 6
Physical frame 0: 2
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3: 3
Physical frame 4:[6]
Page faults: Yes.
Victim frames: 1

Snapshot at step 13:
Program called virtual frame # 5
Physical frame 0: 2
Physical frame 1: 4
Physical frame 2:[5]
Physical frame 3: 3
Physical frame 4: 6
Page faults: No.
Victim frames: None.

Snapshot at step 14:
Program called virtual frame # 3
Physical frame 0: 2
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3:[3]
Physical frame 4: 6
Page faults: No.
Victim frames: None.
```

```
Snapshot at step 15:
Program called virtual frame # 4
Physical frame 0: 2
Physical frame 1:[4]
Physical frame 2: 5
Physical frame 3: 3
Physical frame 4: 6
Page faults: No.
Victim frames: None.

Snapshot at step 16:
Program called virtual frame # 7
Physical frame 0:[7]
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3: 3
Physical frame 4: 6
Page faults: Yes.
Victim frames: 2

Snapshot at step 17:
Program called virtual frame # 3
Physical frame 0: 7
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3:[3]
Physical frame 4: 6
Page faults: No.
Victim frames: None.

Snapshot at step 18:
Program called virtual frame # 5
Physical frame 0: 7
Physical frame 1: 4
Physical frame 2:[5]
Physical frame 3: 3
Physical frame 4: 6
Page faults: No.
Victim frames: None.

Snapshot at step 19:
Program called virtual frame # 6
Physical frame 0: 7
Physical frame 1: 4
Physical frame 2: 5
Physical frame 3: 3
Physical frame 4:[6]
Page faults: No.
Victim frames: None.
Simluation finished. Press enter to continue.
```

Option 6

```
6
Memory Information:
Algorithm Type: LRU
Length of Refernce String: 20
Number of Virtual Pages: 10
Number of Physical Pages: 5
---
Press enter to step through snapshots of physical memory
Or, enter "q" at any time to return to main menu.

Snapshot at step 1:
Program called virtual frame # 0
Physical frame 0:[0]
Physical frame 1:x
Physical frame 2:x
Physical frame 3:x
Physical frame 4:x
Page faults: Yes.
Victim frames: None.

Snapshot at step 2:
Program called virtual frame # 1
Physical frame 0: 0
Physical frame 1:[1]
Physical frame 2:x
Physical frame 3:x
Physical frame 4:x
Page faults: Yes.
Victim frames: None.

Snapshot at step 3:
Program called virtual frame # 2
Physical frame 0: 0
Physical frame 1: 1
Physical frame 2:[2]
Physical frame 3:x
Physical frame 4:x
Page faults: Yes.
Victim frames: None.

Snapshot at step 4:
Program called virtual frame # 3
Physical frame 0: 0
Physical frame 1: 1
Physical frame 2: 2
Physical frame 3:[3]
Physical frame 4:x
Page faults: Yes.
Victim frames: None.

Snapshot at step 5:
Program called virtual frame # 4
Physical frame 0: 0
Physical frame 1: 1
Physical frame 2: 2
Physical frame 3: 3
Physical frame 4:[4]
Page faults: Yes.
Victim frames: None.

Snapshot at step 6:
Program called virtual frame # 5
Physical frame 0:[5]
Physical frame 1: 1
Physical frame 2: 2
Physical frame 3: 3
Physical frame 4: 4
Page faults: Yes.
Victim frames: 0

Snapshot at step 14:
Program called virtual frame # 8
Physical frame 0: 0
Physical frame 1: 1
Physical frame 2: 7
Physical frame 3:[8]
Physical frame 4: 9
Page faults: No.
Victim frames: None.

Snapshot at step 15:
Program called virtual frame # 2
Physical frame 0: 0
Physical frame 1: 1
Physical frame 2:[2]
Physical frame 3: 8
Physical frame 4: 9
Page faults: Yes.
Victim frames: 7

Snapshot at step 16:
Program called virtual frame # 7
Physical frame 0:[7]
Physical frame 1: 1
Physical frame 2: 2
Physical frame 3: 8
Physical frame 4: 9
Page faults: Yes.
Victim frames: 0

Snapshot at step 17:
Program called virtual frame # 3
Physical frame 0: 7
Physical frame 1: 1
Physical frame 2: 2
Physical frame 3: 8
Physical frame 4:[3]
Page faults: Yes.
Victim frames: 9

Snapshot at step 18:
Program called virtual frame # 6
Physical frame 0: 7
Physical frame 1:[6]
Physical frame 2: 2
Physical frame 3: 8
Physical frame 4: 3
Page faults: Yes.
Victim frames: 1

Snapshot at step 19:
Program called virtual frame # 4
Physical frame 0: 7
Physical frame 1: 6
Physical frame 2: 2
Physical frame 3:[4]
Physical frame 4: 3
Page faults: Yes.
Victim frames: 8

Snapshot at step 20:
Program called virtual frame # 5
Physical frame 0: 7
Physical frame 1: 6
Physical frame 2:[5]
Physical frame 3: 4
Physical frame 4: 3
Page faults: Yes.
Victim frames: 2
Simluation finished. Press enter to continue.
```